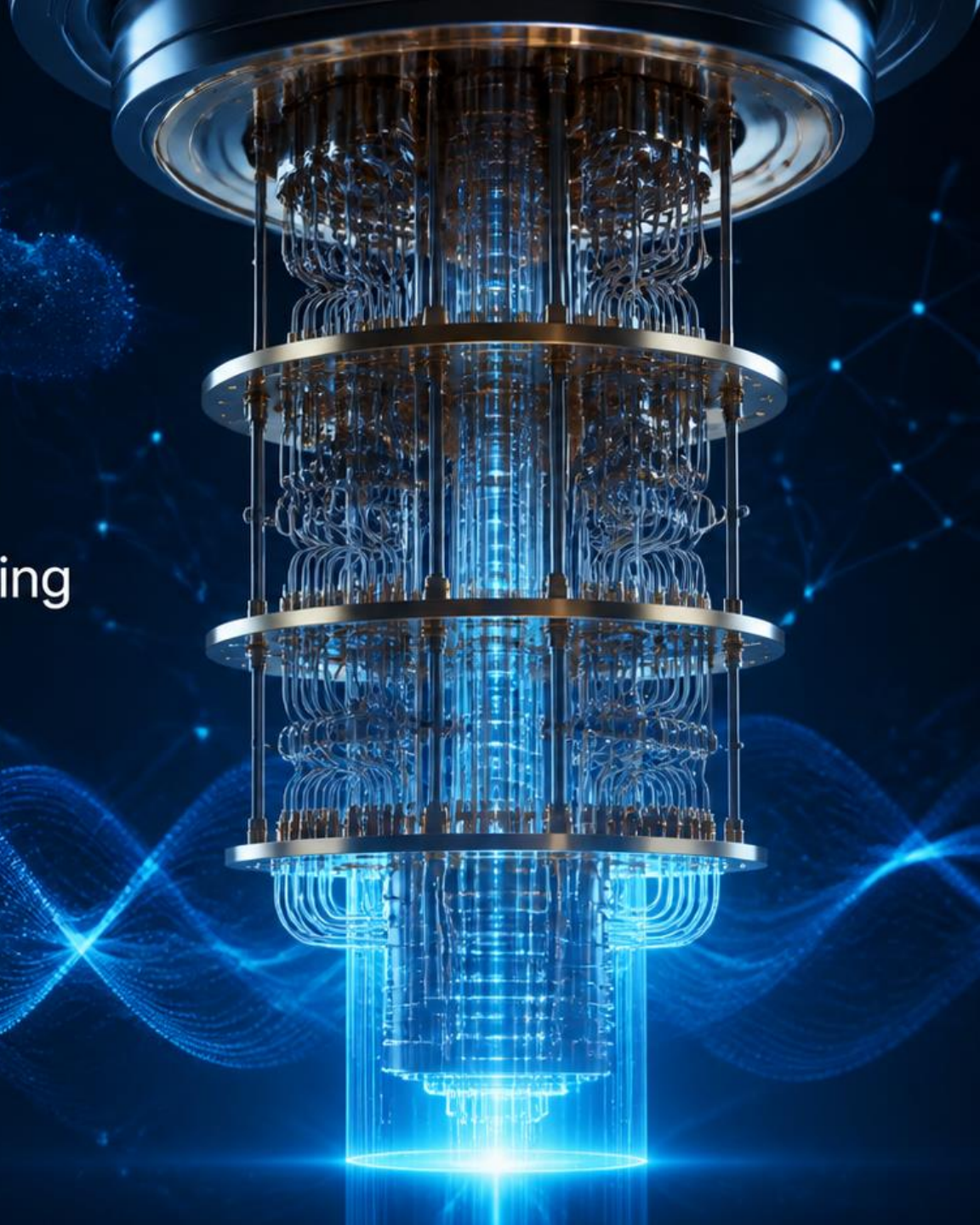
NIC26

A Brave New World

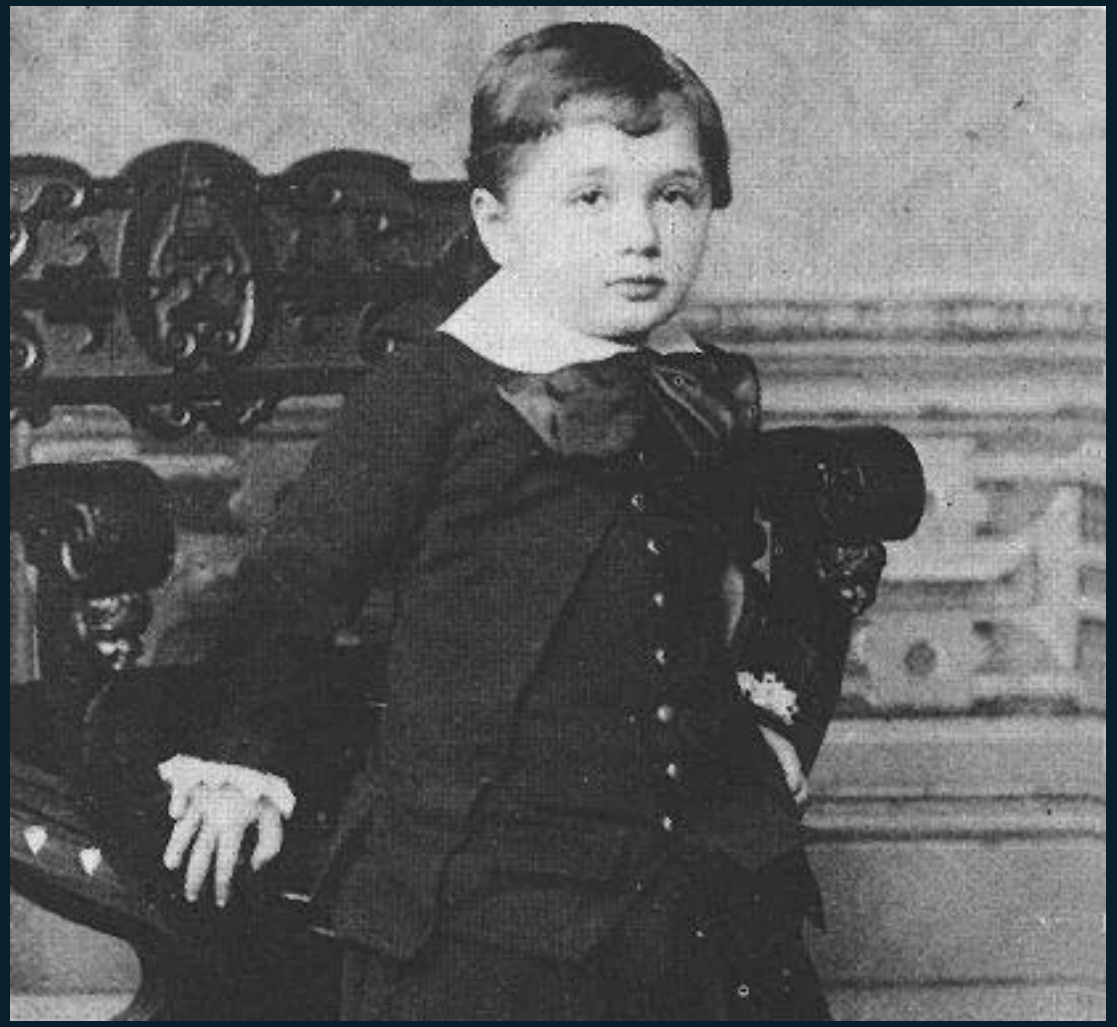
A Technical Introduction to Quantum Computing



Andy Malone MVP
Andy@Andymalone.org


Now, a New Era Begins!

Albert Einstein



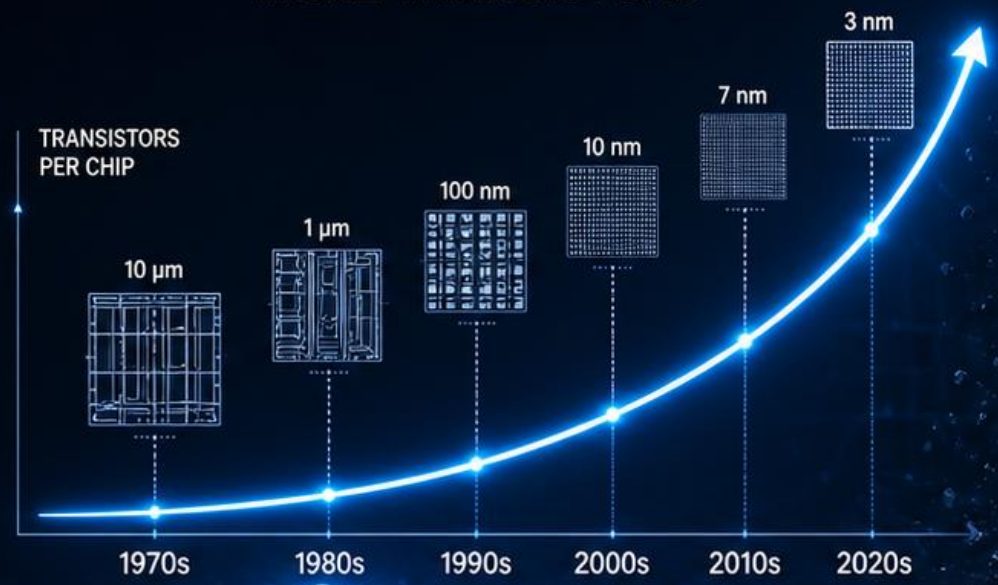




But Why Does the World Needs Quantum Now?

BEYOND MOORE'S LAW

MORE TRANSISTORS

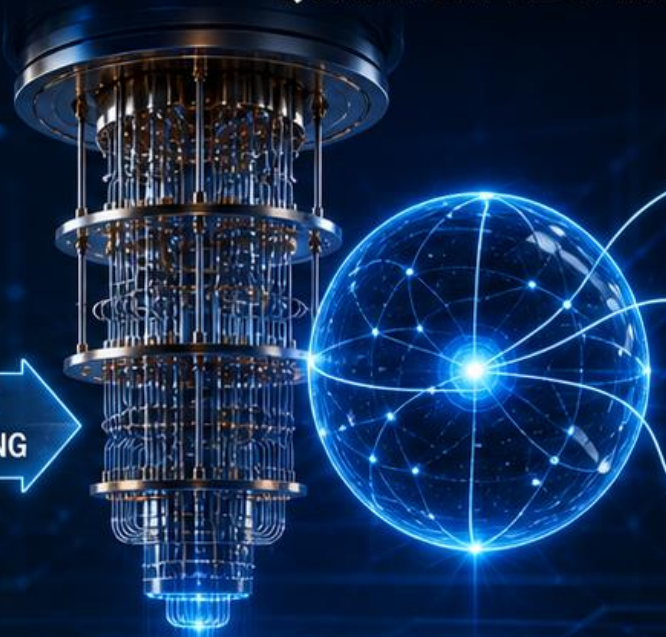


PHYSICAL + ECONOMIC LIMITS

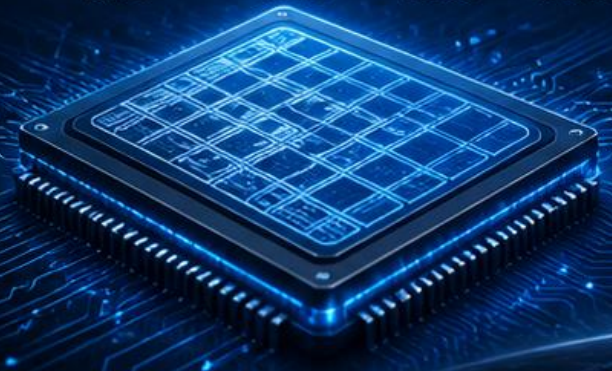


HYBRID COMPUTING

QUANTUM ADVANTAGE



- CHEMISTRY
- MATERIALS
- OPTIMIZATION





Quantum accelerates selected problems — it does not replace every computer.

INSIDE A QUANTUM COMPUTER

Main components of a superconducting quantum computer



Classical computer

Runs programs, coordinates operations and processes results.



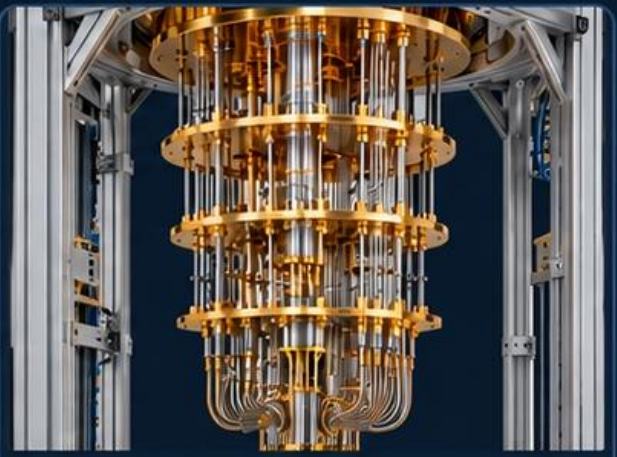
Control electronics

Creates precise microwave pulses to control the qubits.



Cables and filters

Carry signals to the chip while reducing unwanted noise.



Dilution refrigerator

Cools the quantum chip to just above absolute zero.



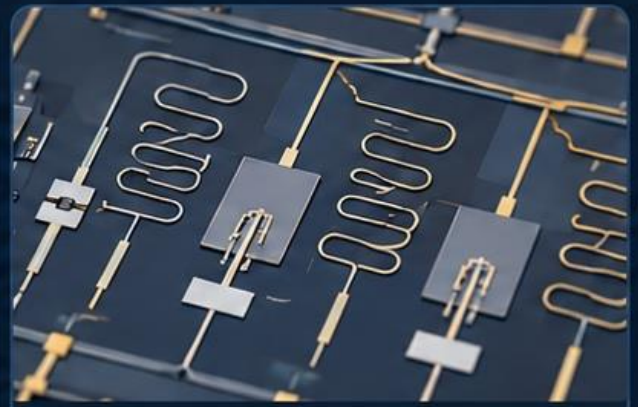
Shielding and enclosure

Protect the chip from heat, stray light and magnetic fields.



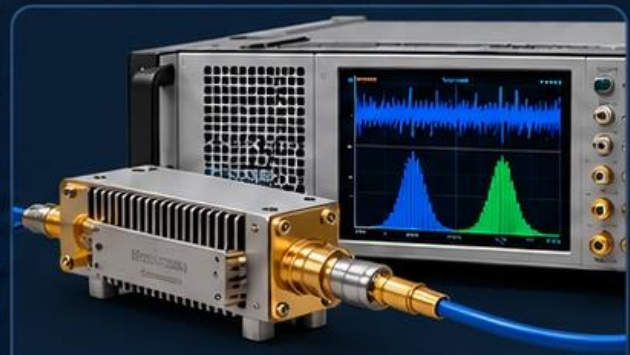
Quantum processor

Contains qubits that store and process quantum information.



Resonators and couplers

Help read qubits and enable interactions between them.



Readout electronics

Amplifies and measures signals to produce classical results.

Simplified overview • Illustrations not to scale • Other quantum computer designs use different hardware

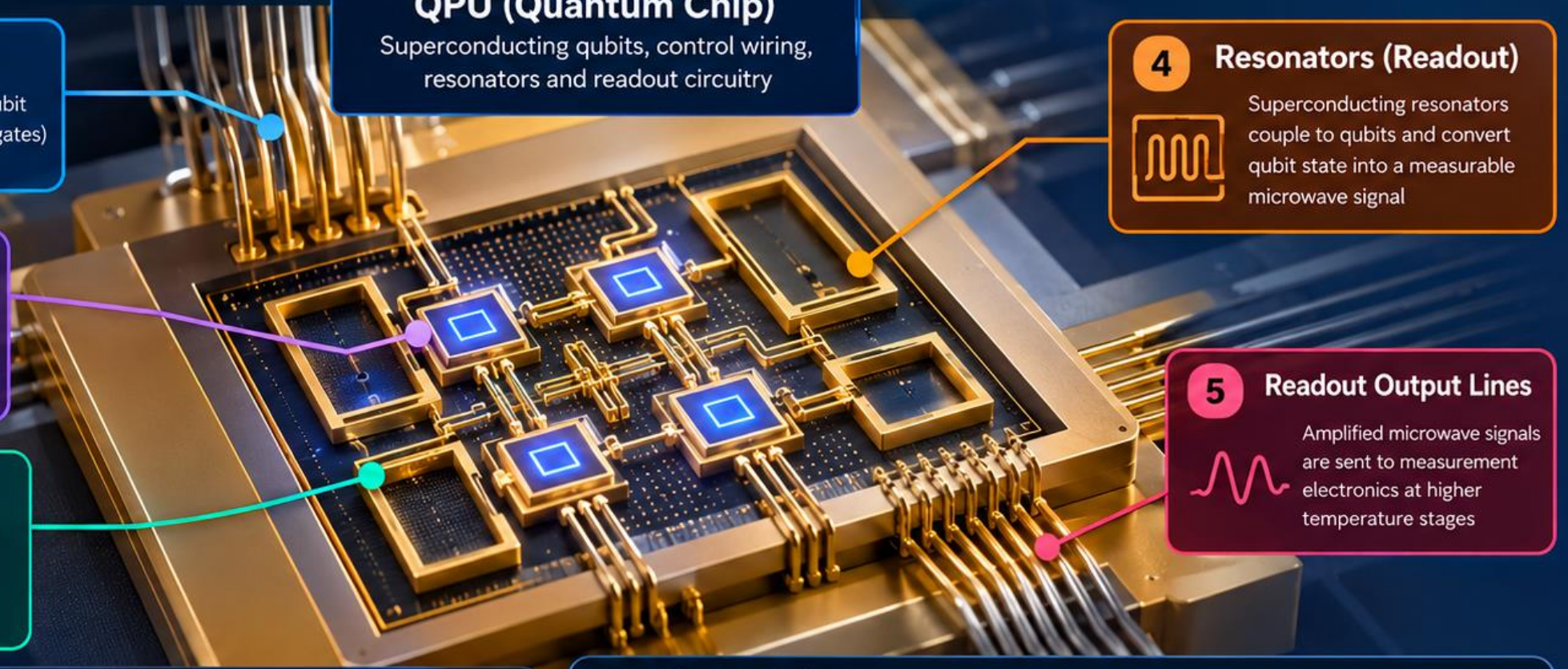
QPU Architecture

Inside a quantum processing unit

A QPU manipulates and measures qubits using precisely timed microwave or laser signals at ultra-low temperatures, with integrated control wiring, resonators and readout.

QPU (Quantum Chip)

Superconducting qubits, control wiring, resonators and readout circuitry



1 Control Input Lines

Microwave signals control qubit state (single- and two-qubit gates)

2 Qubits

Superconducting qubits (e.g. transmons) store quantum information

3 Coupling Elements

Tunable couplers allow qubits to interact for two-qubit gates (entanglement)

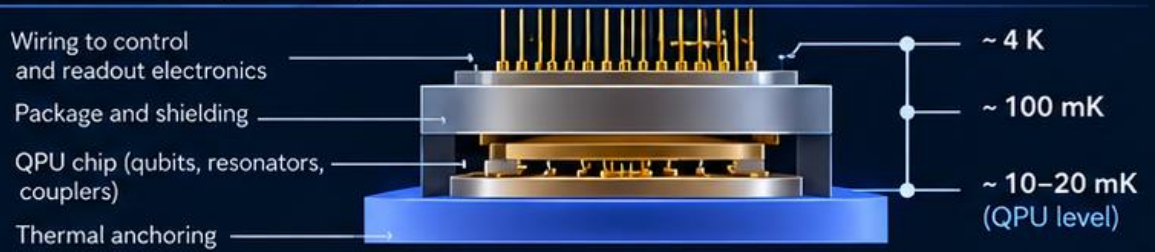
4 Resonators (Readout)

Superconducting resonators couple to qubits and convert qubit state into a measurable microwave signal

5 Readout Output Lines

Amplified microwave signals are sent to measurement electronics at higher temperature stages

Cross-section (side view)



Key Characteristics

Operates at ~10–20 mK
Superconducting circuitry requires ultra-low temperatures

Microwave control
Precise, timed pulses implement quantum gates

Integrated resonators & couplers
Enable measurement and qubit interactions

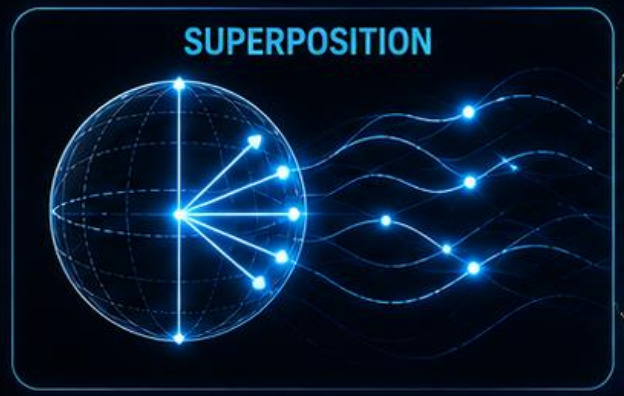
Scalable architecture
Many qubits with on-chip control, readout and routing

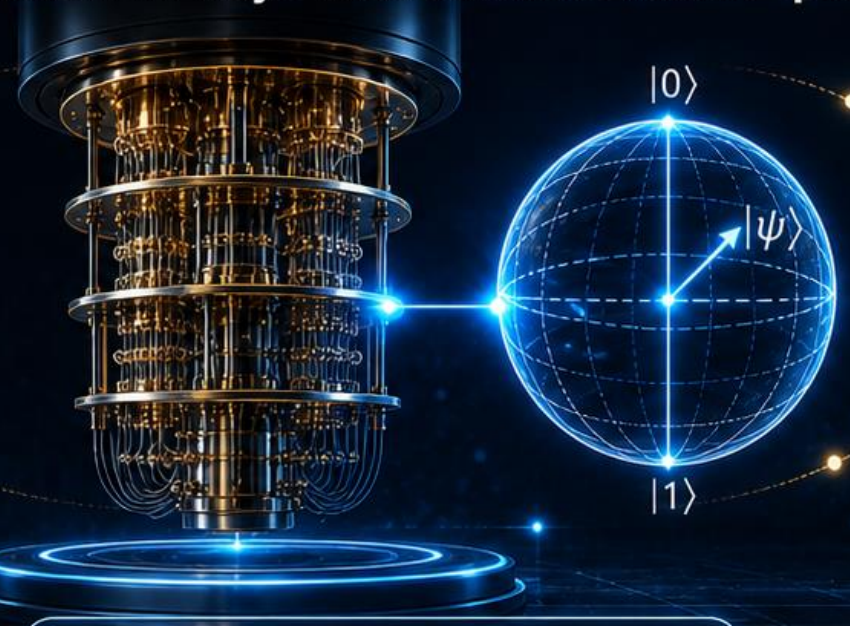


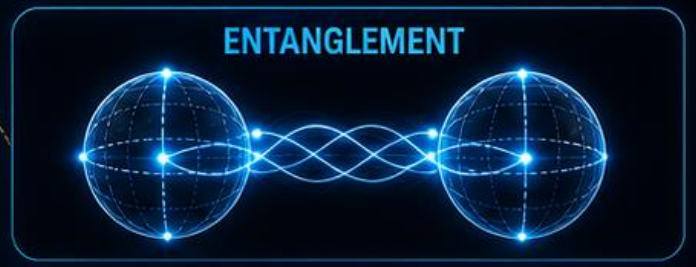
But Andy, How Does Quantum Work?

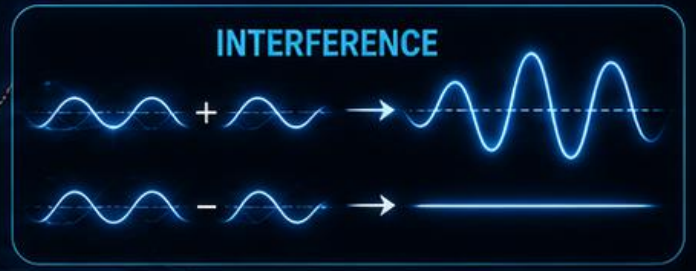
WHAT IS QUANTUM COMPUTING?

Quantum computing uses qubits and quantum effects —
superposition, entanglement and interference —
to process information in ways that can accelerate specific problems.











CLASSICAL INPUT



QUANTUM PROCESSING



CLASSICAL RESULT

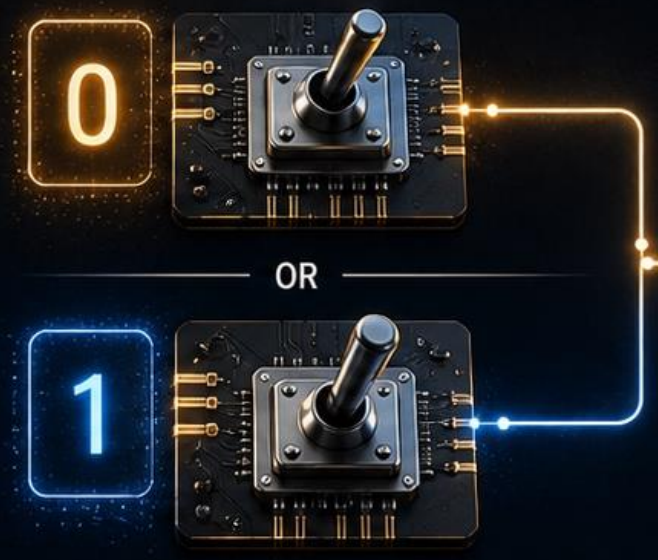
Best suited to selected problems — not a replacement for every computer.

CHEMISTRY • MATERIALS • OPTIMISATION • CRYPTOGRAPHY

BITS vs QUBITS

CLASSICAL BIT

0 OR 1



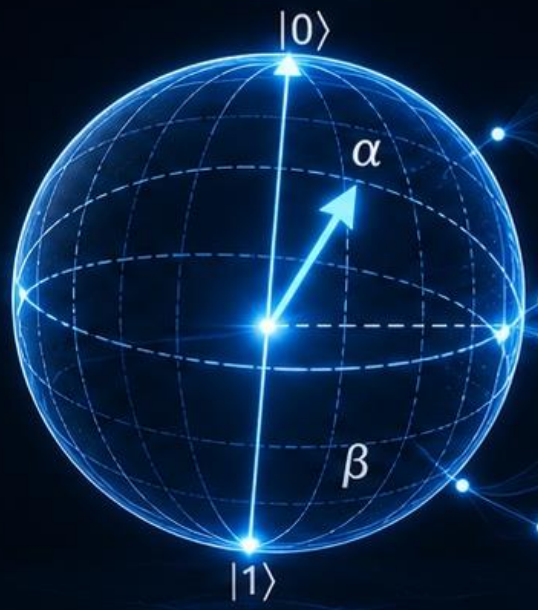
DETERMINISTIC STATE

FROM STATE TO AMPLITUDE



QUBIT

$\alpha|0\rangle + \beta|1\rangle$



SUPERPOSITION

INTERFERENCE

ENTANGLEMENT



MEASUREMENT → 0 OR 1

n bits: one value



→ one n-bit value at a time

A qubit is not simply both answers at once.

n qubits: 2^n amplitudes



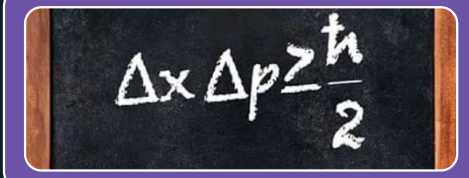
→ describes amplitudes across 2^n basis states

Quantum Mechanics 5 Core Concepts



Wave-Particle Duality

- Quantum objects, like electrons, can exhibit both wave-like and particle-like behavior depending on how they are observed



Uncertainty Principle

- It is impossible to precisely measure certain properties of a quantum object, such as position and momentum, simultaneously.



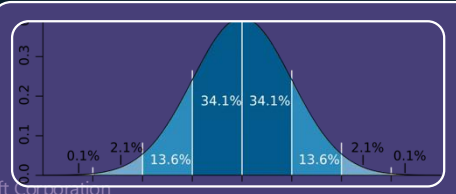
Superposition

- Quantum objects can exist in multiple states simultaneously, which is known as a superposition of states



Entanglement

- Quantum objects can become “entangled” in a way that their properties are correlated, even when separated by large distances. E.g. Quantum Teleportation.



Probabilistic Nature

- Quantum mechanics is based on probabilities rather than definite outcomes, as measured by the wave function and the Schrödinger equation

©Microsoft Corporation
Azure

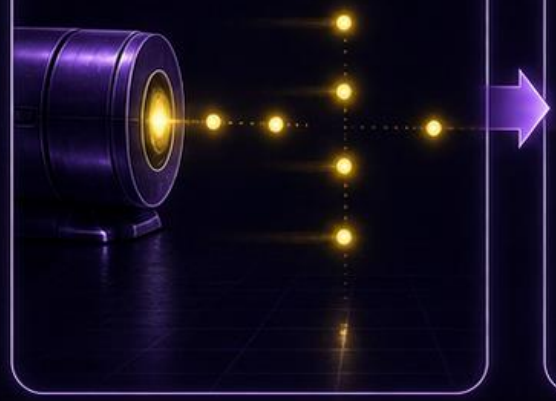


Wave Particle Duality

WAVE–PARTICLE DUALITY

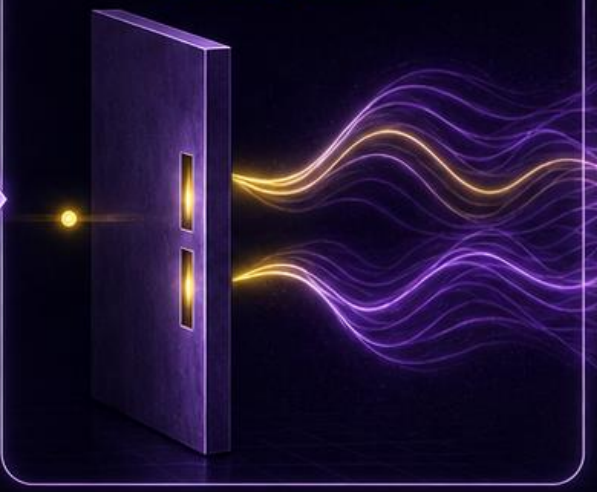
1 ONE QUANTUM AT A TIME

Detected as a particle



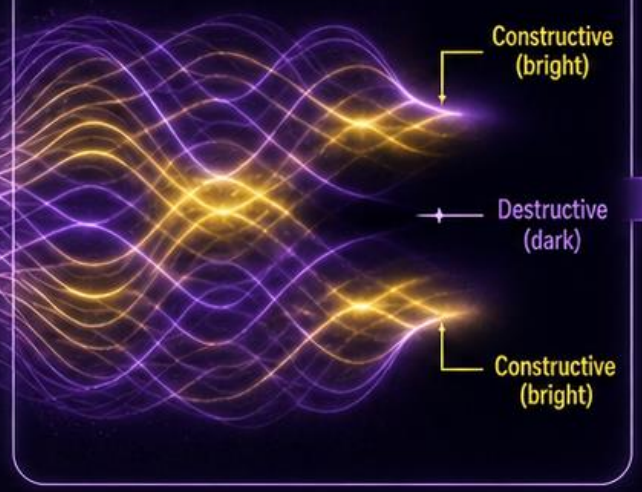
2 TWO POSSIBLE PATHS

Probability amplitudes pass through both slits



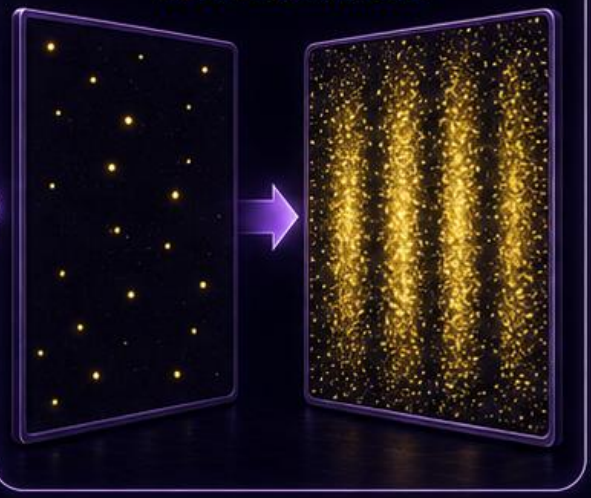
3 INTERFERENCE

Amplitudes reinforce or cancel



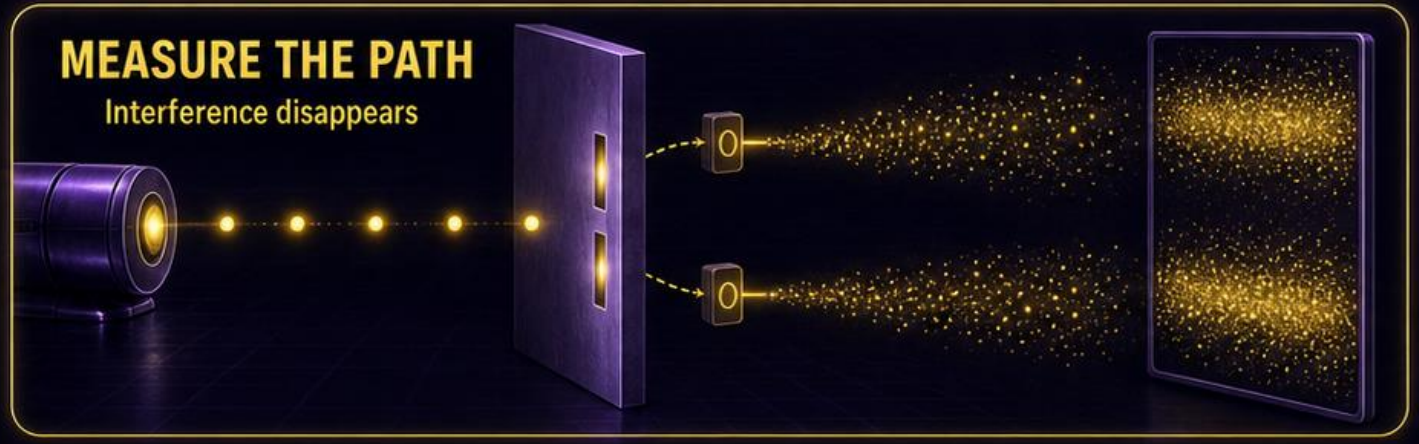
4 PATTERN EMERGES

Individual impacts build a wave-like distribution



MEASURE THE PATH

Interference disappears





Quantum objects are described by a **wavefunction** — but every detection is a **discrete event**.



It is **not switching** between a tiny ball and a physical water wave.



The Uncertainty Principle

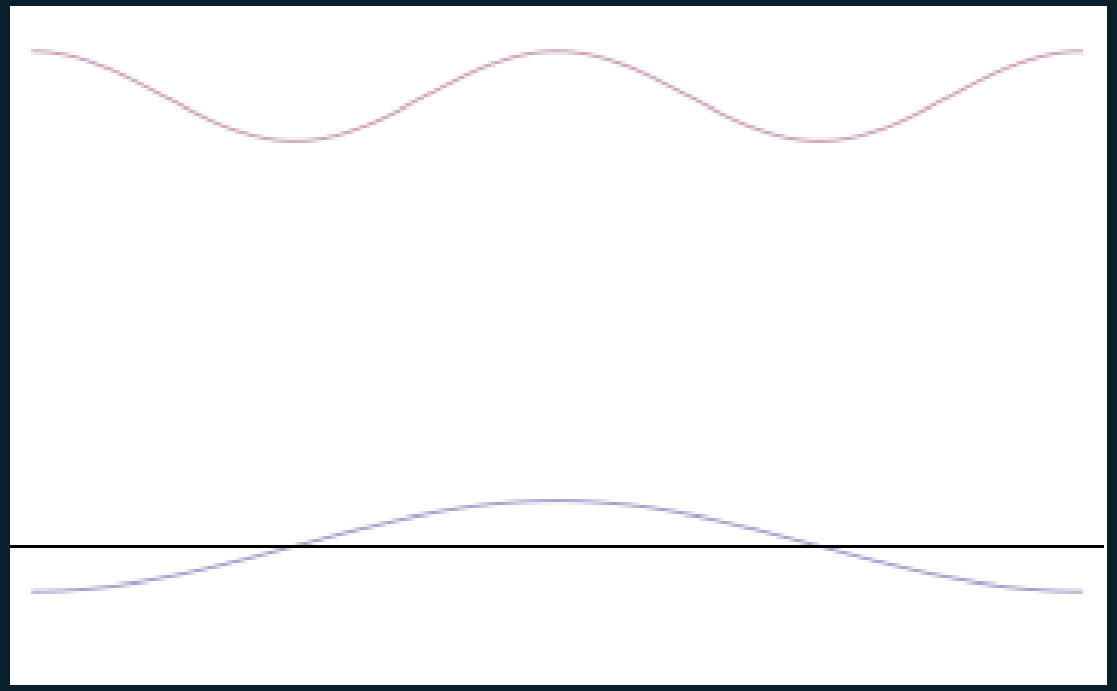
Heisenberg Uncertainty Principle



- States that in quantum mechanics,
- **Position** and **Momentum** cannot both be known with perfect accuracy at the same time.
  - The more we measure one, the less we can know the other.
- This isn’t due to technological limits,
- **It's a fundamental property of nature**.
  - Quantum particles don’t have exact values for both simultaneously; they exist in a range of possibilities until measured.

Heisenberg Uncertainty Principle

- Reality is probabilistic rather than deterministic.
- This insight directly influences quantum cryptography, where unpredictability and measurement disturbance are used to detect eavesdropping.



The superposition of several plane waves to form a wave packet. The waves shown here are real for illustrative purposes only; in quantum mechanics the wave function is generally complex.



Superposition

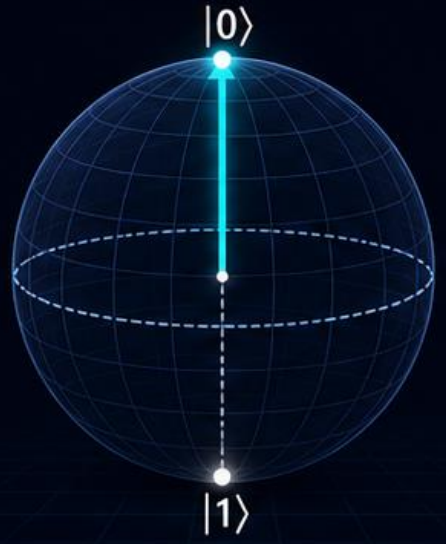
We can be in many places at the same time! The Multiverse

SUPERPOSITION

A qubit is a controlled combination of basis states

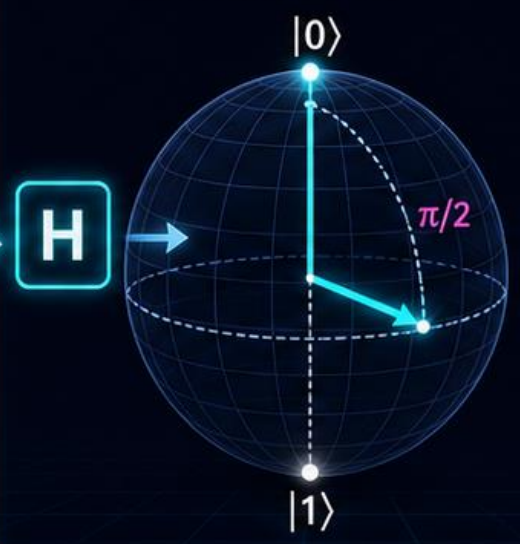
1 START WITH $|0\rangle$

Known basis state



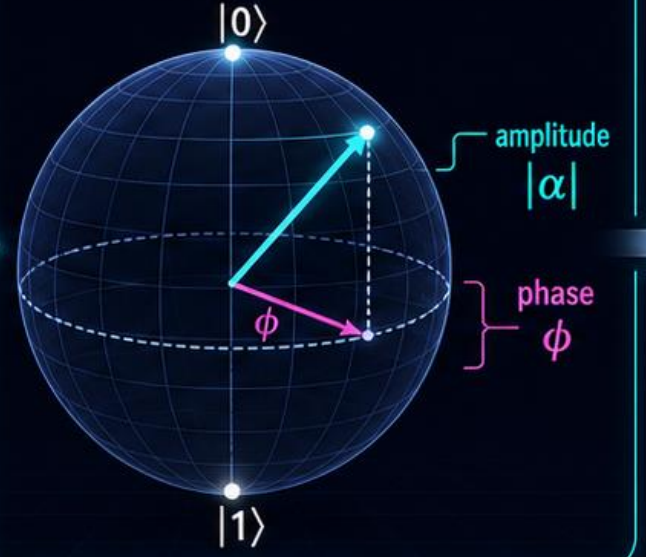
2 APPLY A GATE

H rotates the state



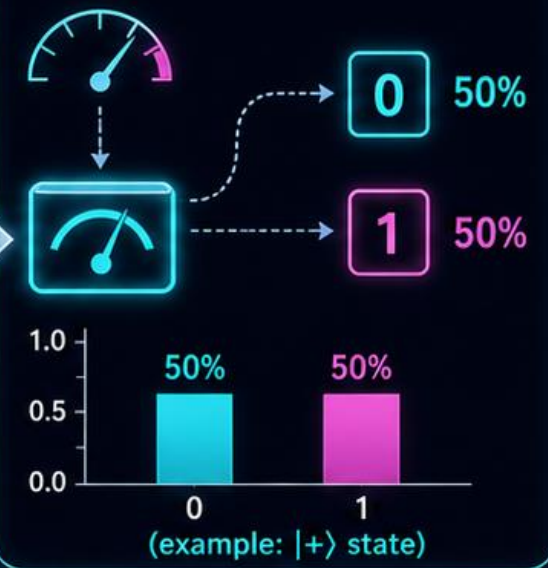
3 $\alpha|0\rangle + \beta|1\rangle$

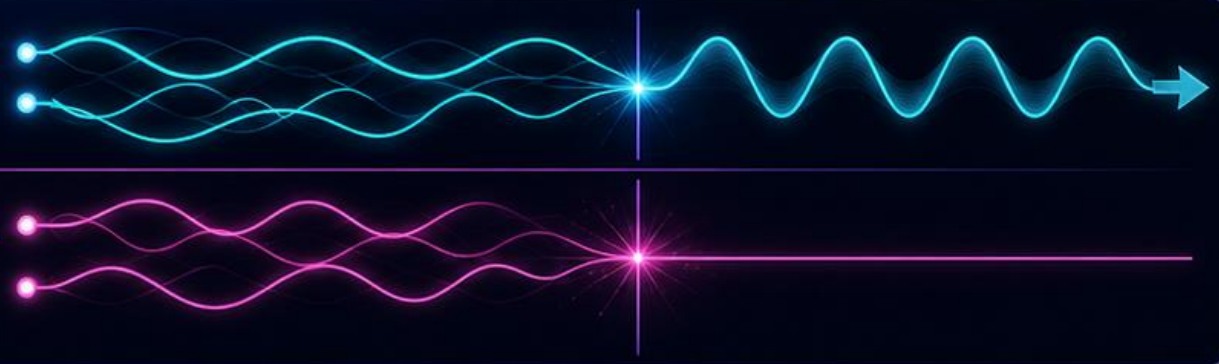
Amplitudes carry probability + phase



4 MEASURE

One result per shot: 0 or 1





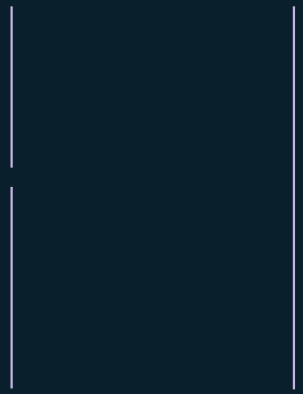
Superposition becomes useful through interference.

NOT: two classical answers stored and readable at once

Another Example



Photons



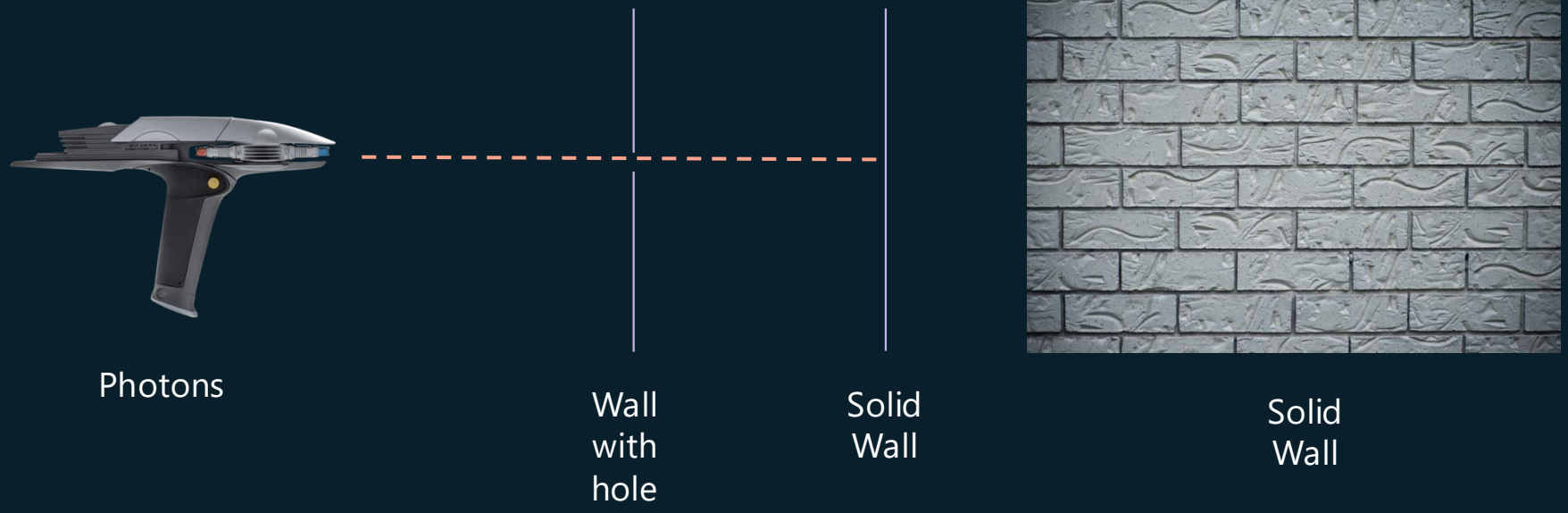
Wall with hole

Solid Wall

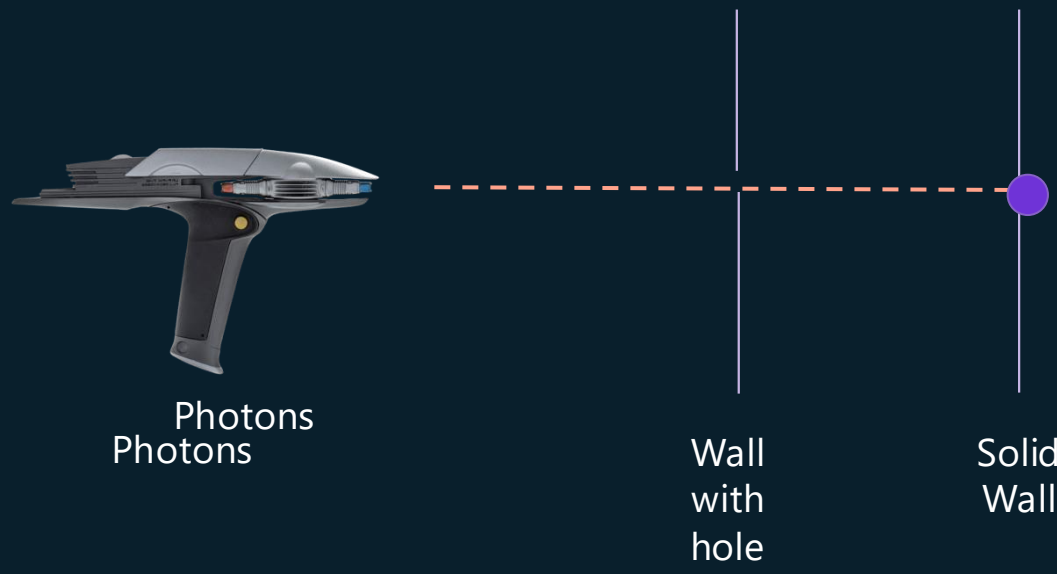


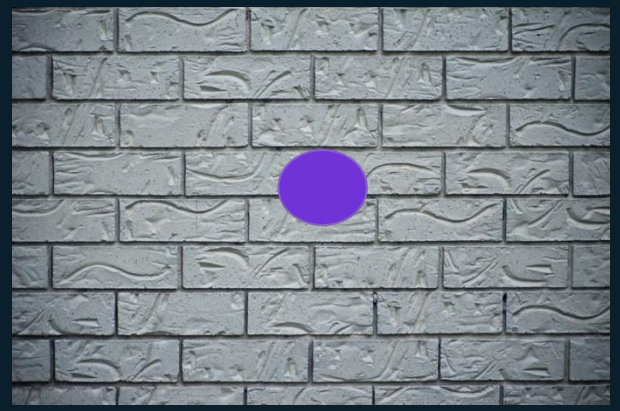
Solid Wall

Superposition Illustrated



Superposition Illustrated





Superposition Illustrated



Photons



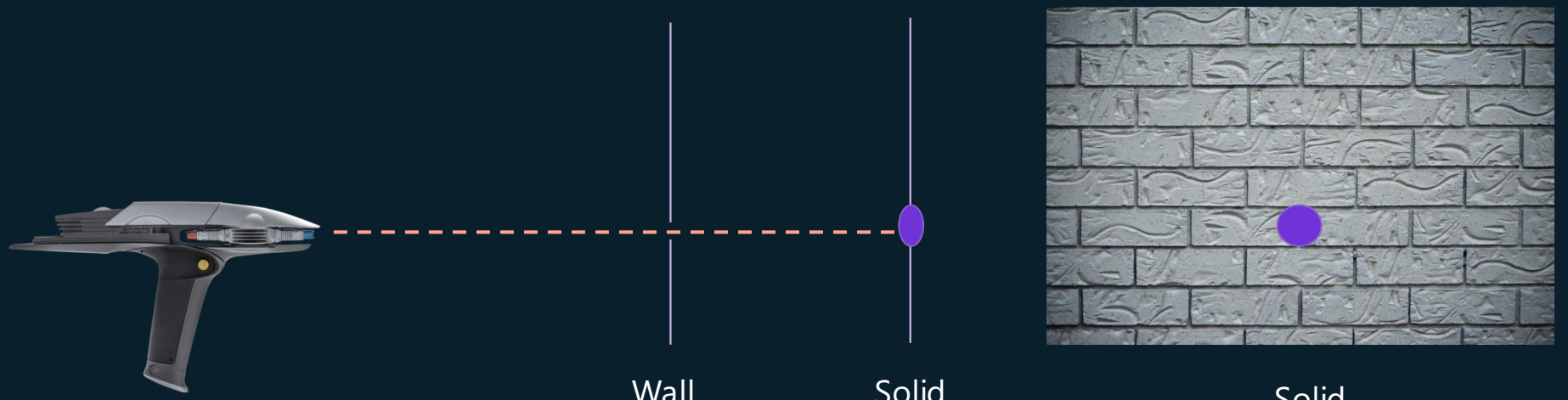
Wall with new hole

Solid Wall



Solid Wall

Superposition Illustrated



Repeated over time this demonstrates a binary process 1 or 0

Superposition Illustrated



Photons

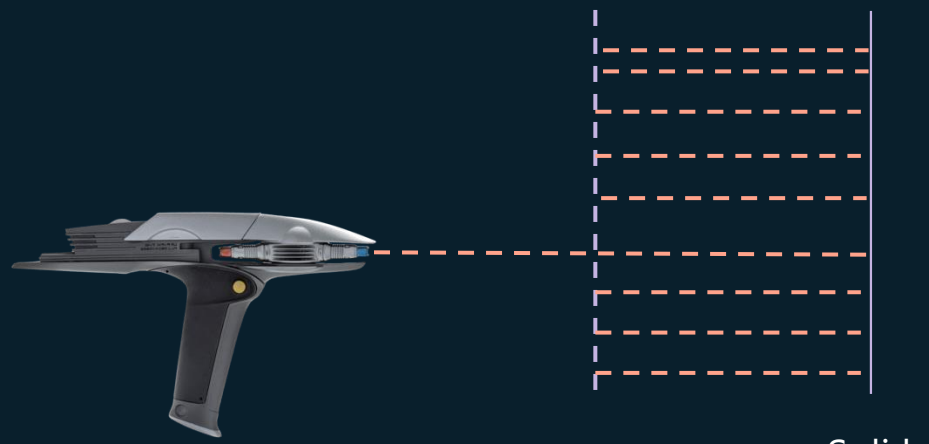
Wall with many holes

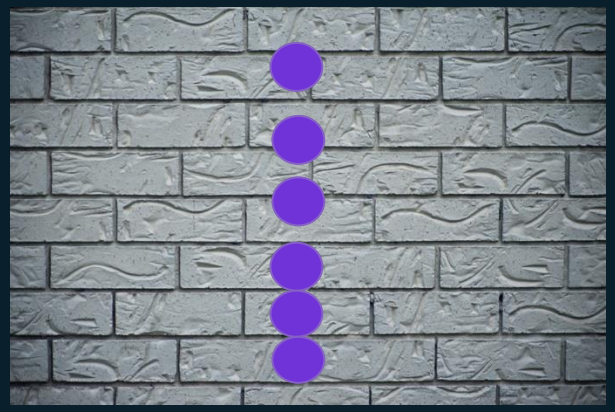
Solid Wall



Solid Wall

Superposition Illustrated





Superposition of States

Now, 1 Single Photon Becomes Many!

So, 1 Key could now open 1000 boxes!!

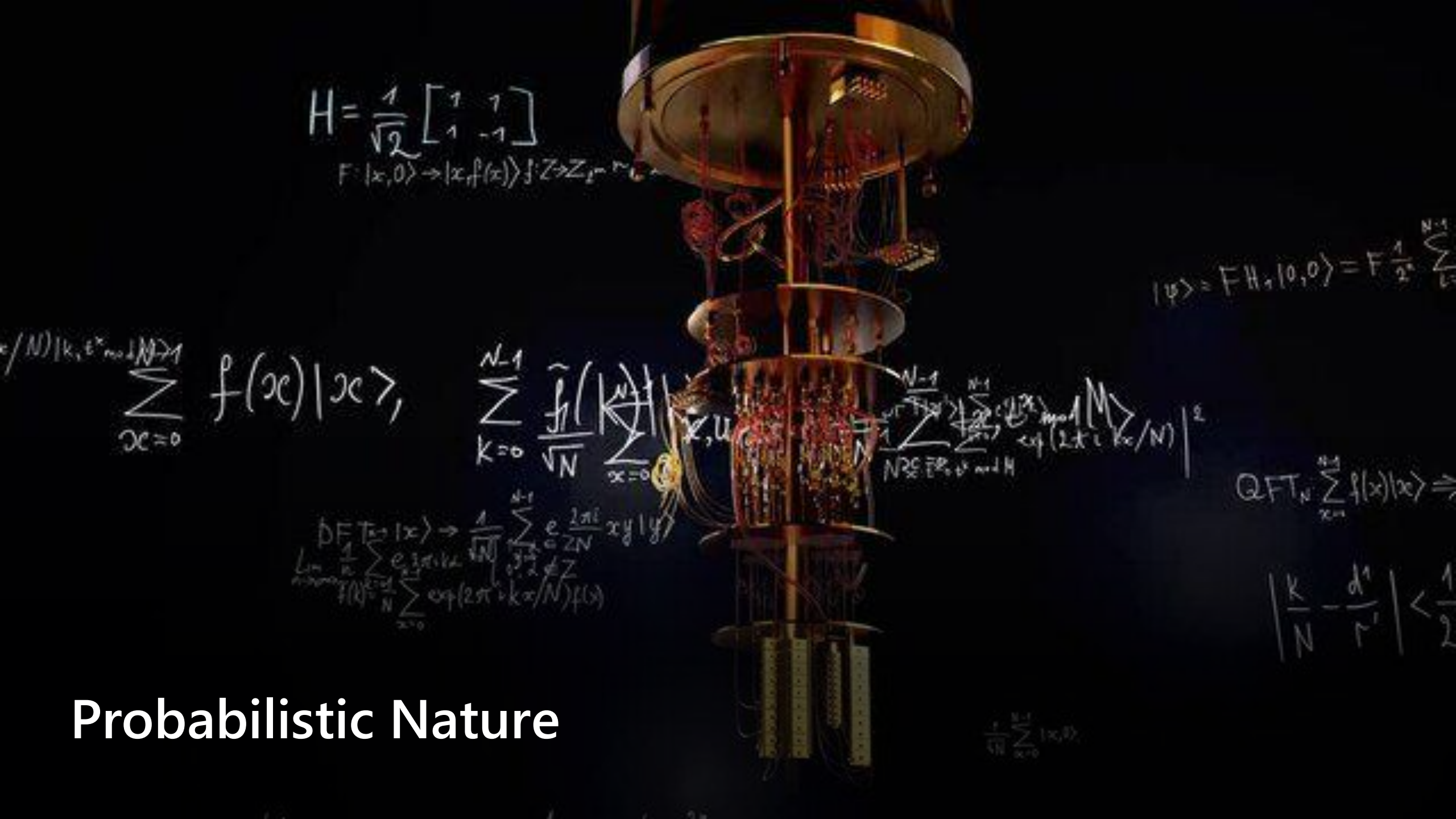


Entanglement

Entanglement

- **Entanglement**: Quantum objects can become "entangled" in a way that their properties are correlated, even when separated by large distances. E.g. Two Spinning Coins.
- If **superposition** is *one object being in multiple states at once*
- **Then Entanglement** is when **two or more objects share a *linked* state**, no matter how far apart they are.
- Once particles become entangled:
  - Measuring one **instantly affects** the state of the other,
  - Even if they're **light-years apart**.





Probabilistic Nature

Probabilistic Nature

- Quantum mechanics is based on probabilities rather than definite outcomes, as measured by the wave function and the Schrödinger equation.
  - There are two main interpretations:
    - **Nature is fundamentally random.**
    - **Reality is deterministic but hidden (hidden variables)**
    - We just don’t have access to everything — the "randomness" is fake.

Classical World

- If you know the initial conditions (e.g., position & velocity), You can predict the Outcome.

Quantum World

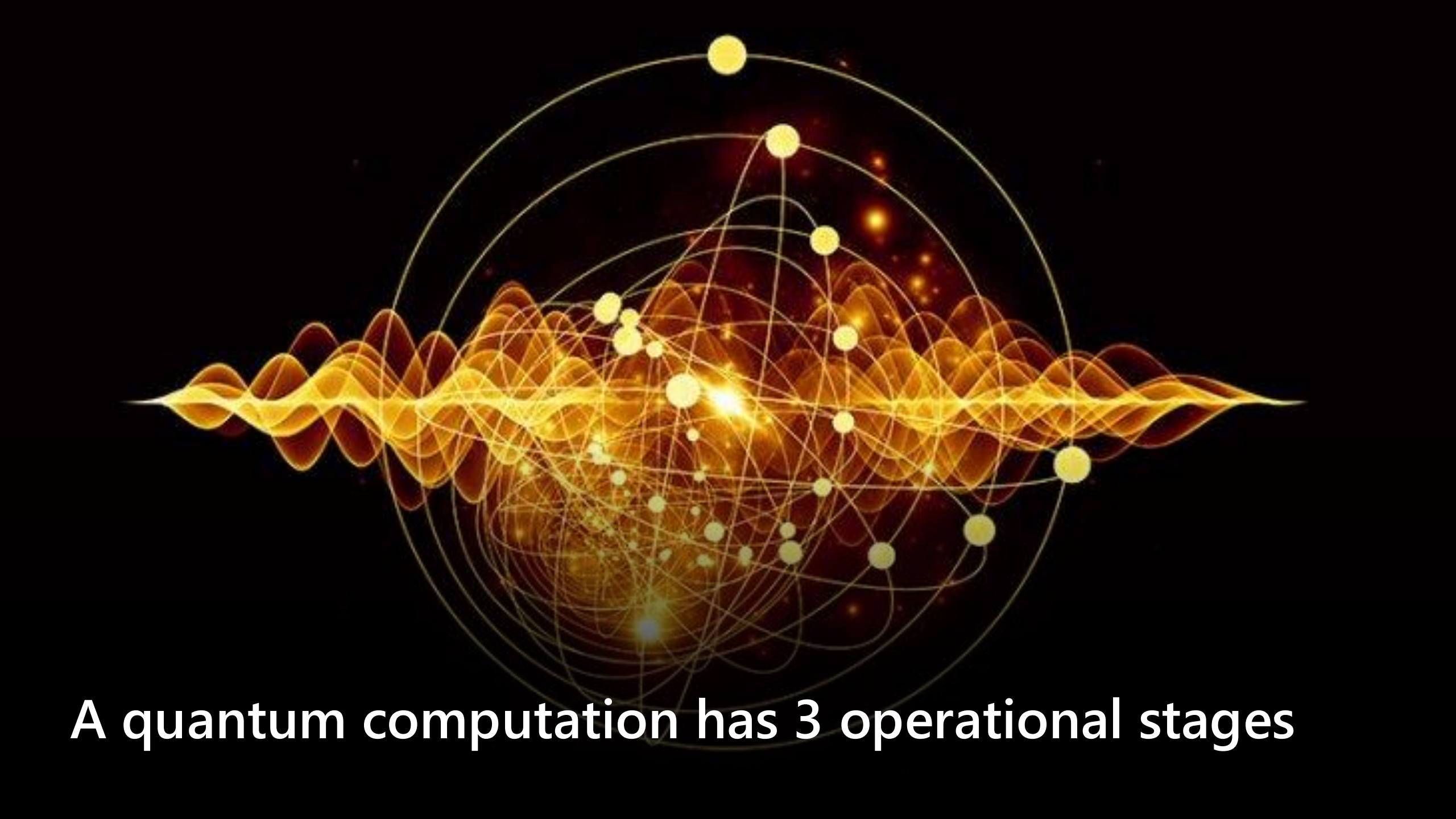
- Even if you know everything about a quantum system, you can only predict probabilities of outcomes.

Probabilistic Nature - Schrödinger’s cat



$$|\psi\rangle = 21(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

- Imagine **two boxes**, each with a Schrödinger’s-style cat. The cats are **entangled**, so:
- They are **not individually** alive or dead.
- Instead, the *pair* are in a shared state like:
  Both alive *or* **both dead,** but you don’t know which until one is observed.
- The moment you **open one box** and see “alive,” you *instantly* know the other must **also be alive**, even if it's on Mars.

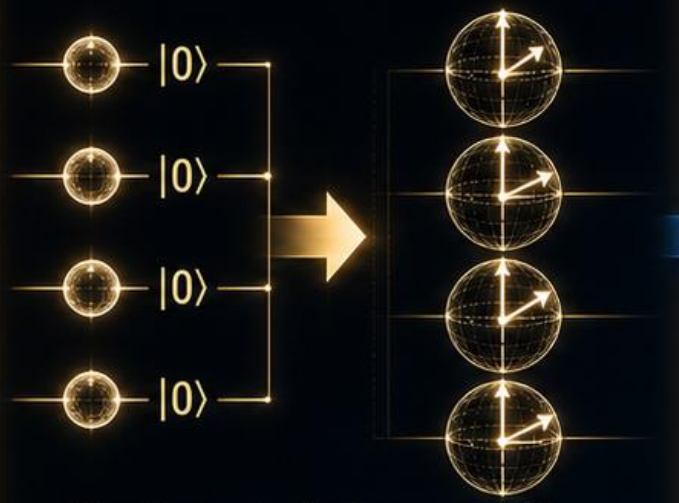


A quantum computation has 3 operational stages

THE THREE STAGES OF QUANTUM COMPUTATION

1 INITIALIZE

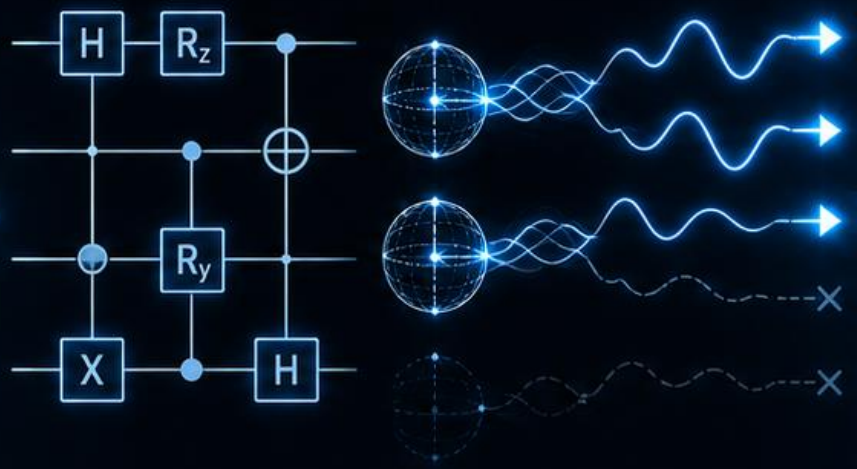
Prepare qubits in a known state



Create controlled superposition

2 MANIPULATE

Quantum gates transform amplitudes



Entanglement + interference shape the result

3 MEASURE

Read qubits as classical 0s and 1s



Repeat shots to estimate probabilities

PREPARE → TRANSFORM → READ

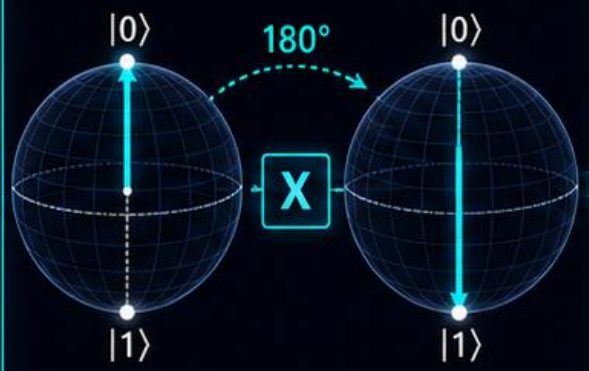
Classical control coordinates every stage.

QUANTUM GATES

Controlled, reversible operations that rotate and correlate qubit states

1 X BIT FLIP

$|0\rangle \leftrightarrow |1\rangle$



Like NOT

2 H HADAMARD

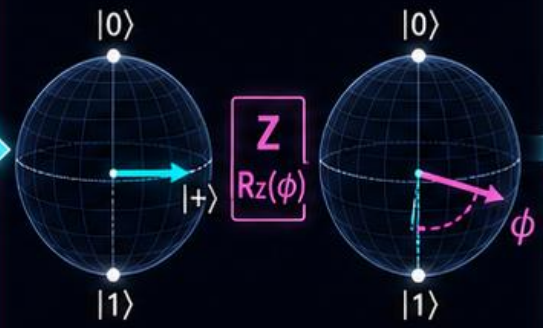
Creates superposition

$|0\rangle \rightarrow |+\rangle$



3 Z / Rz PHASE

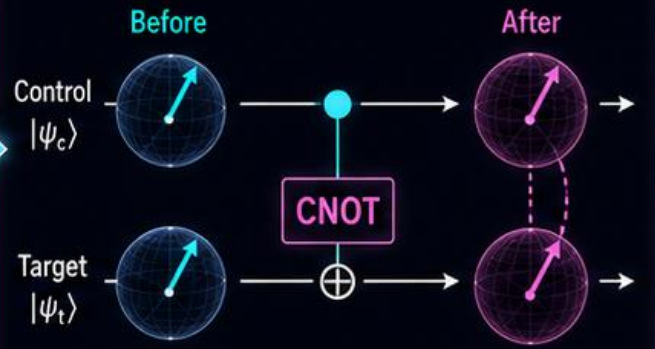
Changes relative phase
Enables interference



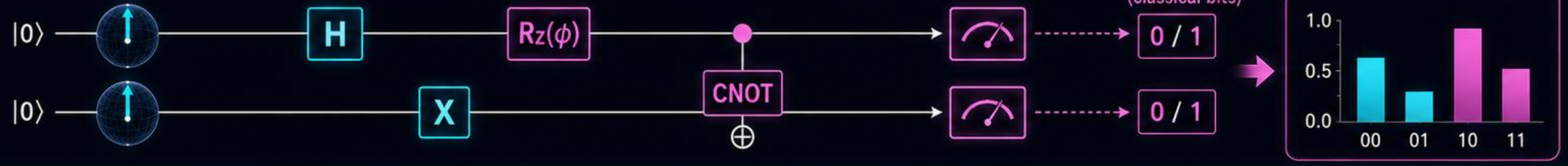
Probabilities may look unchanged until later interference.

4 CNOT CONTROLLED

Changes the target only when control = 1
Can create entanglement



PREPARE → APPLY GATES → MEASURE



A quantum gate is not a physical doorway. It is a precisely controlled transformation.

HOW A QUANTUM CIRCUIT WORKS

A quantum circuit is a timed sequence of operations on qubits

1 COMPILE

Map the algorithm to gates

2 INITIALIZE

Prepare qubits in $|0\rangle$

3 APPLY GATES

Rotate amplitudes + phase

4 ENTANGLE + INTERFERE

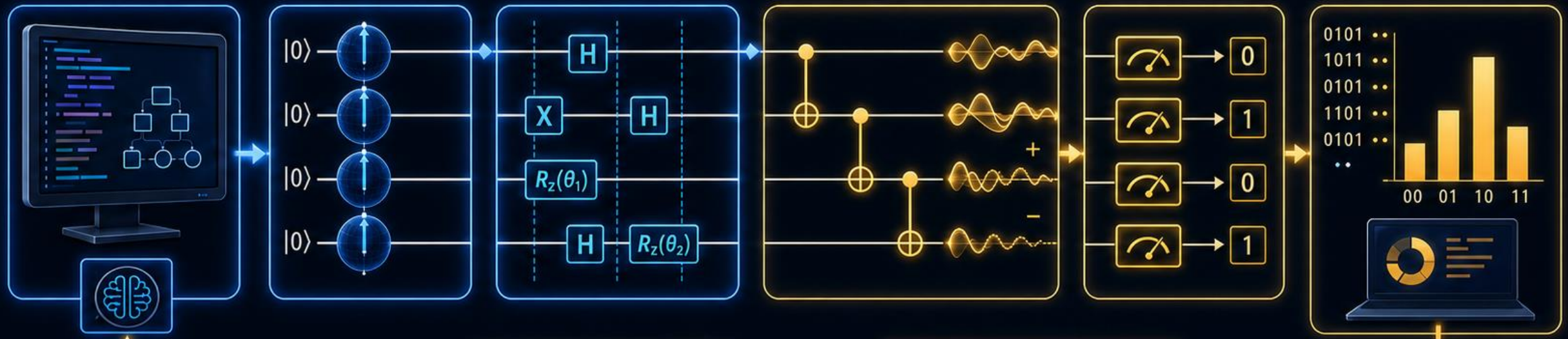
Correlate qubits and shape probabilities

5 MEASURE

Convert qubits to classical bits

6 REPEAT + ANALYSE

Run many shots to build a distribution



UPDATE PARAMETERS

CLASSICAL CONTROL

The circuit shapes probabilities; it does not reveal every possible answer.

Time flows left → right

Three quantum algorithms, simply explained

What they do — and why they matter

Grover

Search faster

Find one target in a large, unsorted set.



Fewer checks

For N possibilities, roughly $\sqrt{N}$ quantum checks instead of N classical checks.

Useful speedup — not an instant answer.

Deutsch-Jozsa

Spot a hidden pattern

Is a rule's output always the same, or split equally between 0 and 1?



One quantum query

Works when these are the only two possible cases.

A demonstration of quantum advantage.

Shor

Find factors efficiently

Break a large number into the numbers that multiply to make it.

$15 \rightarrow 3 \times 5$

Simple example

Major security impact

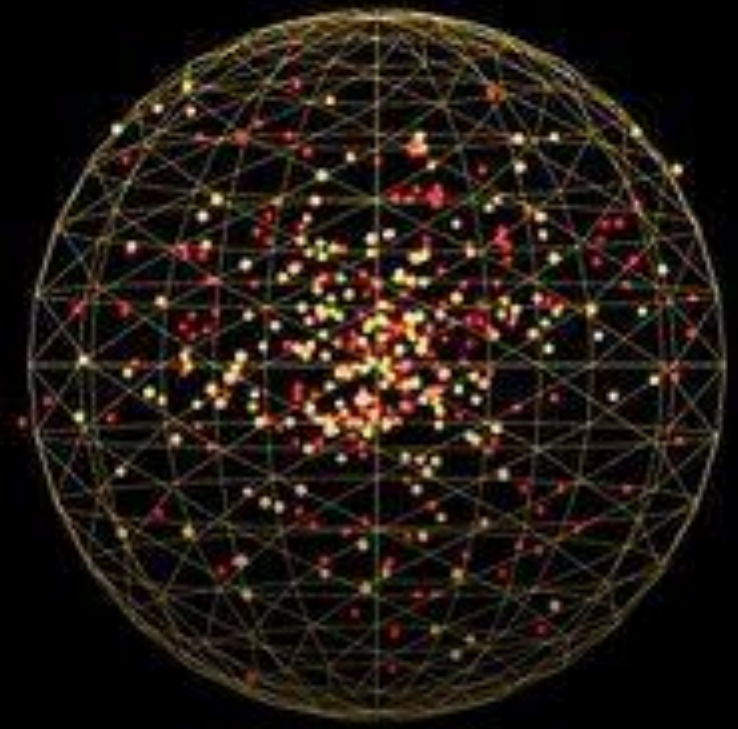
Also solves discrete logarithms, threatening RSA, Diffie–Hellman and elliptic-curve cryptography.

Requires a sufficiently capable quantum computer.

What security teams can do now

1. Find vulnerable cryptography
2. Prioritise data that must stay secret
3. Plan a move to post-quantum cryptography

Encrypted data collected now could be decrypted in the future. Migration takes time.



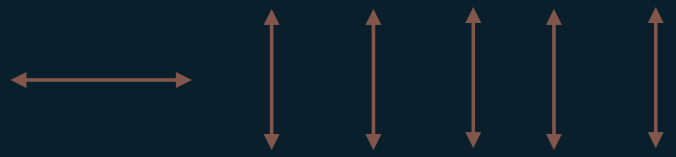
Quantum Key Distribution (QKD)

Quantum Key Distribution (BB84)

1. Draw a random line of Rs & Bs

B R R R B B

2. Draw a sequence of directions



3. Encode 1 & 2 into Qubits



4. Send Qubits to Bob



Alice

1. Bob receives Qubits



2. Draw Random Bases

3. Measure in Random Bases



B R B R B B



Bob

Quantum Key Distribution (BB84)

1. Draw a random line of Rs & Bs

B R R R B B

Note that the outcome is identical, if measurement basis are the same.
Otherwise, they would be identical only with a probability of 50%



Alice

1. Bob receives Qubits



2. Draw Random Bases

3. Measure in Random Bases

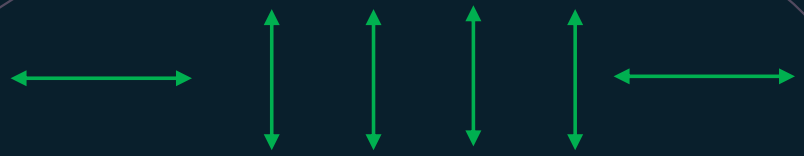
B R B R B B



Bob

Post Processing Steps

B R R R B B



Hi everyone. I used these basis.



Alice

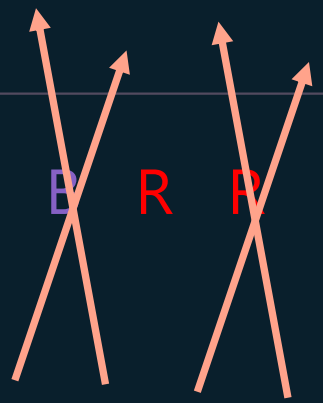
Bob

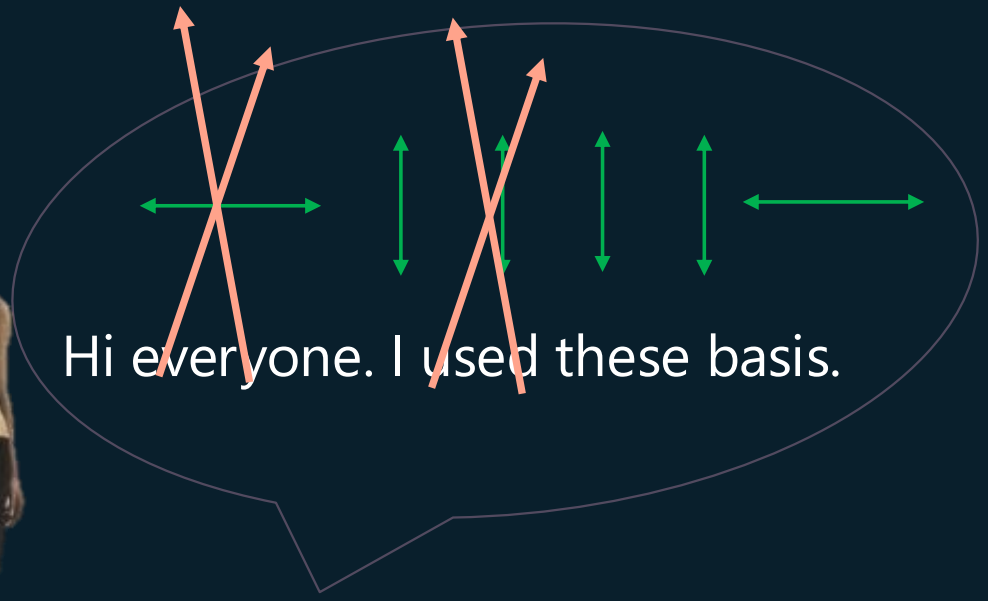


I everyone. I used these basis.

B R B R B B

Post Processing Steps

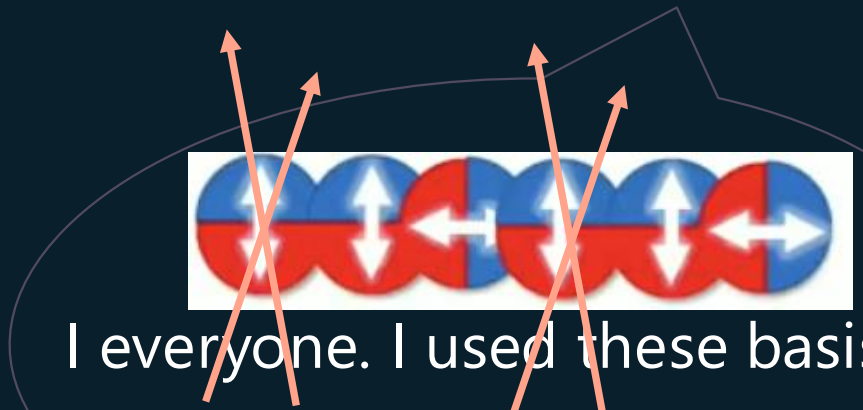


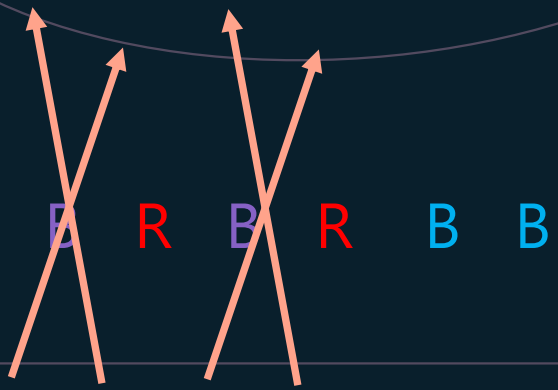




Bob







Post Processing Steps

Key R R B B



Alice

Bob



Key R R B B


NISQ Era:
The Current Reality

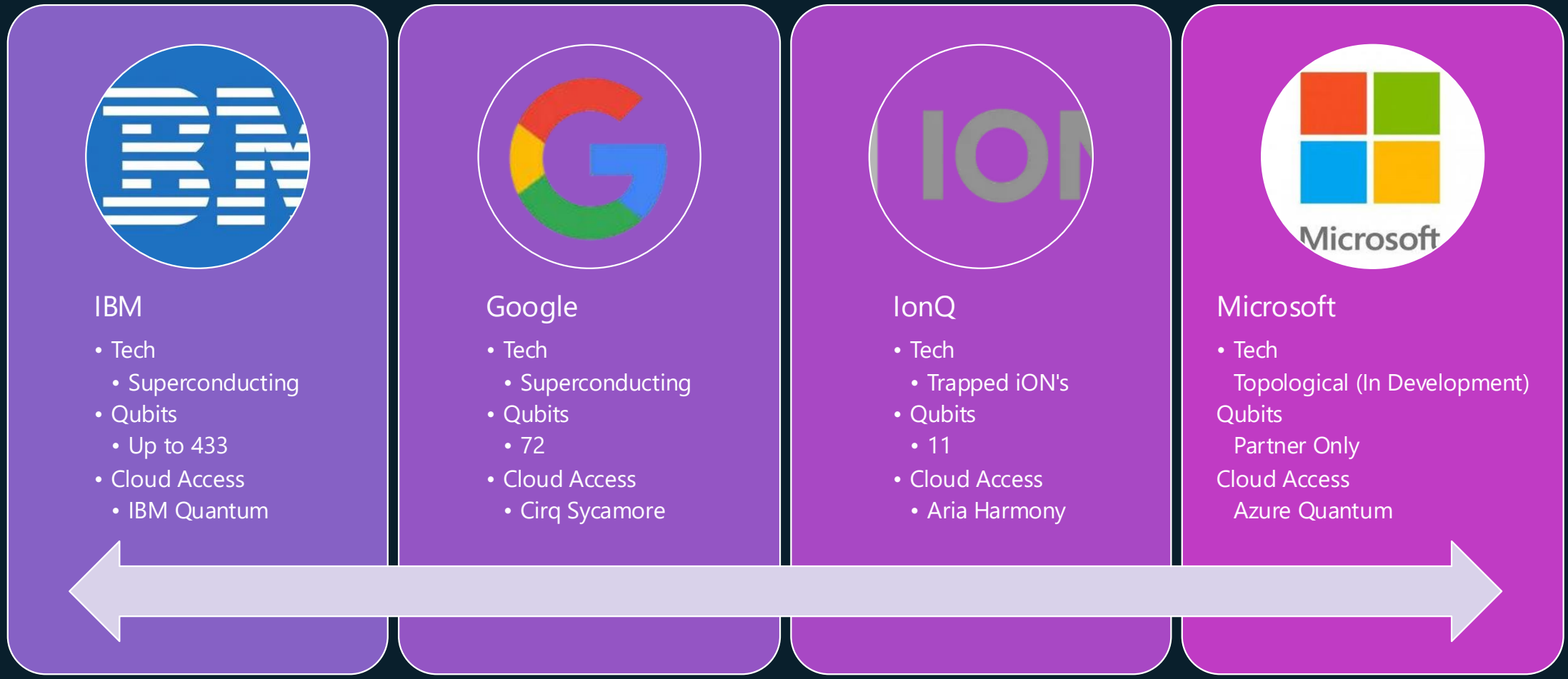
Noisy Intermediate-Scale Quantum (2023-2027 era)

Qubits	Error Rate	Gate Time	Coherence
• 50 -1000	• 0.1%-1%	• 20-200ns	• µs-ms

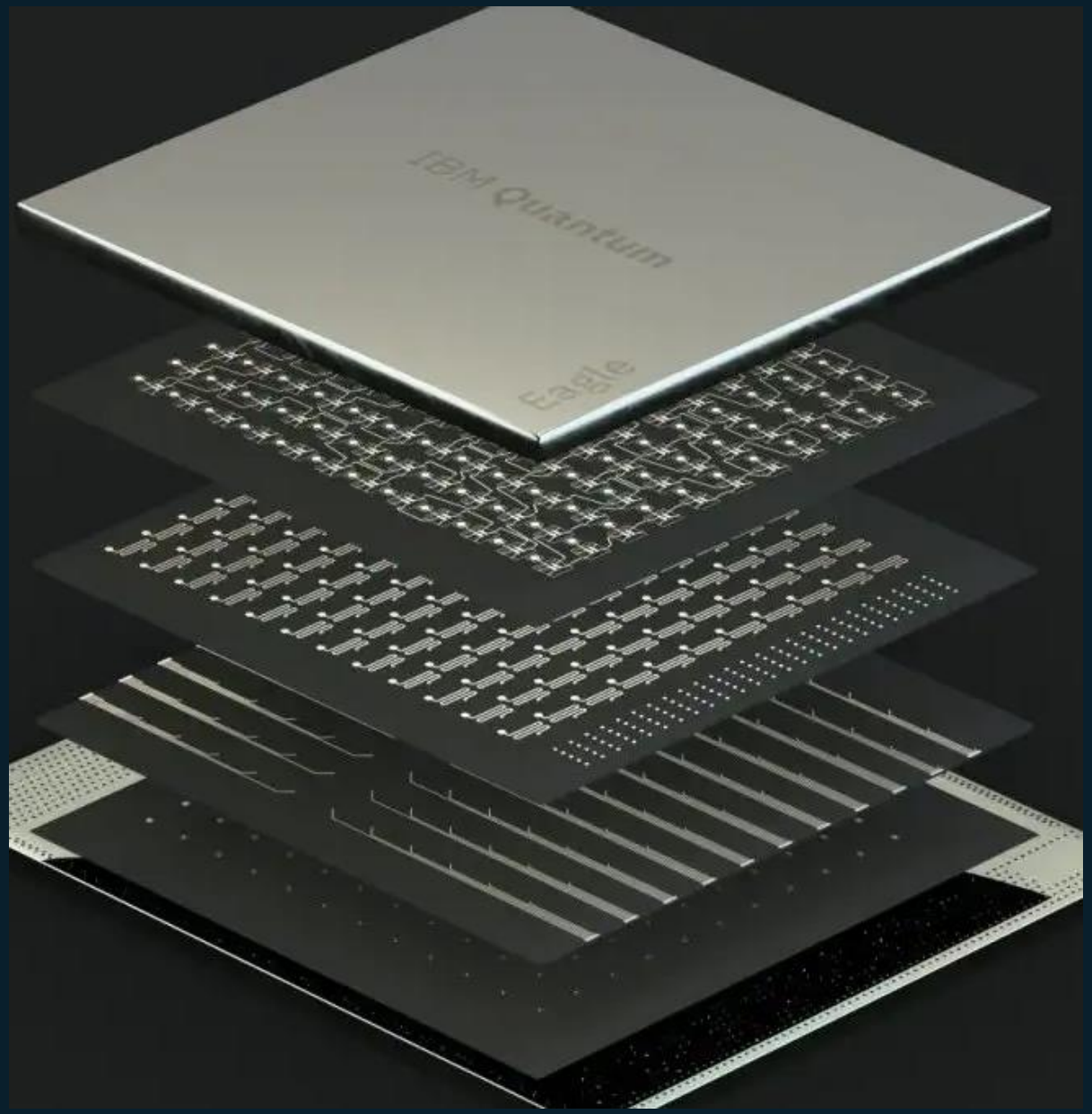
Major Challenges

Decoherence	Gate Errors	Scaling
Environmental noise destroys quantum state	Each operation ~0.1-1% failure rate	1000 physical qubits → 1 logical qubit needed

Current Quantum Platforms



©Microsoft Corporation
Azure



Comparison of QPU Types

- Superconducting (IBM, Google): Fast progress, but noisy
- Trapped ions (IonQ, Honeywell): High fidelity, slow to scale
- Photonic (PsiQuantum): Theoretically scalable, not yet proven
- Topological (Microsoft): Intrinsically stable, but still experimental

Real World Applications (2026 – 2030)



Drug Discovery

Simulate protein folding & molecular interactions



Finance

Portfolio optimization, risk analysis



Cryptography

Factor RSA | Quantum-resistant schemes

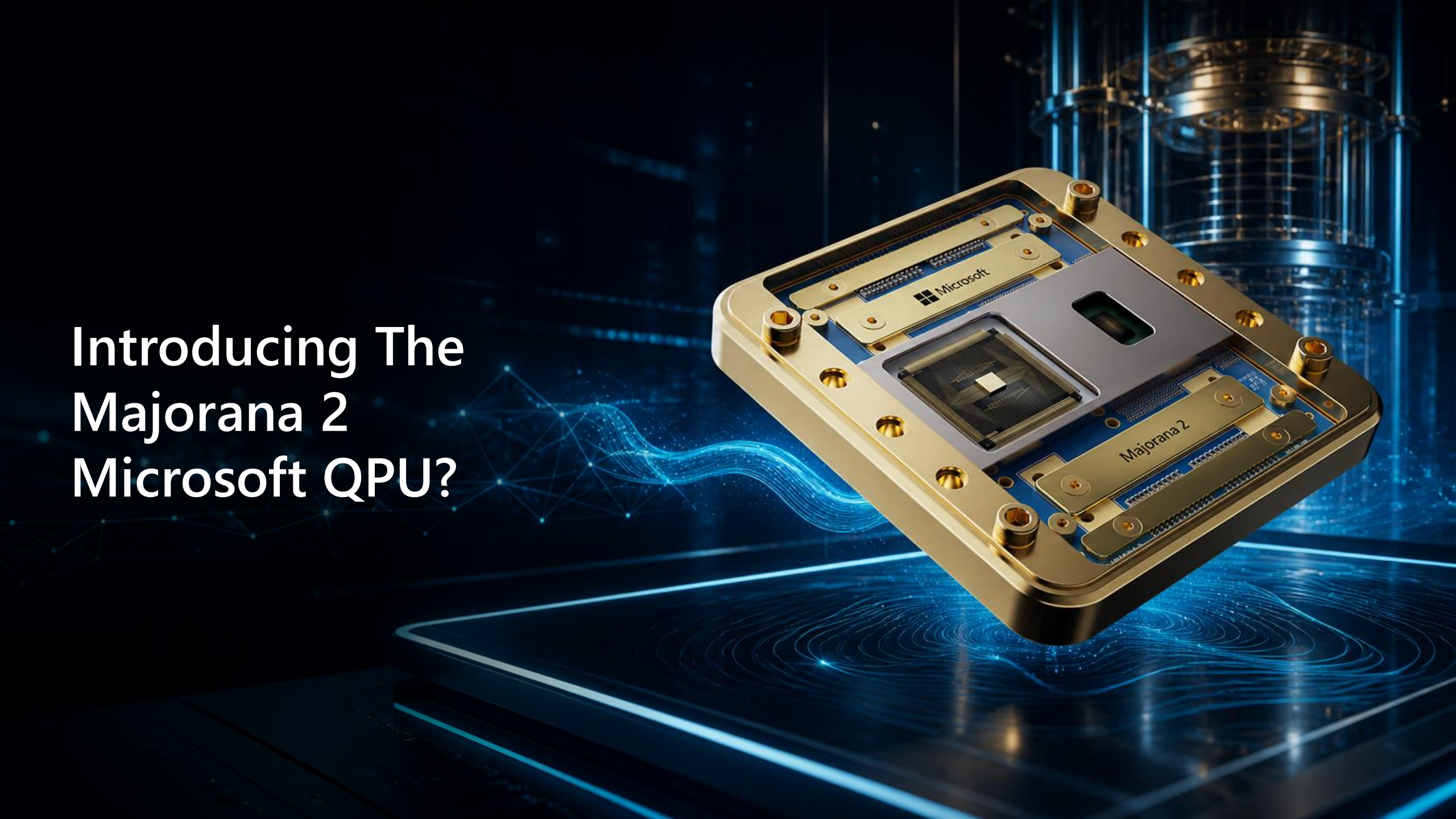


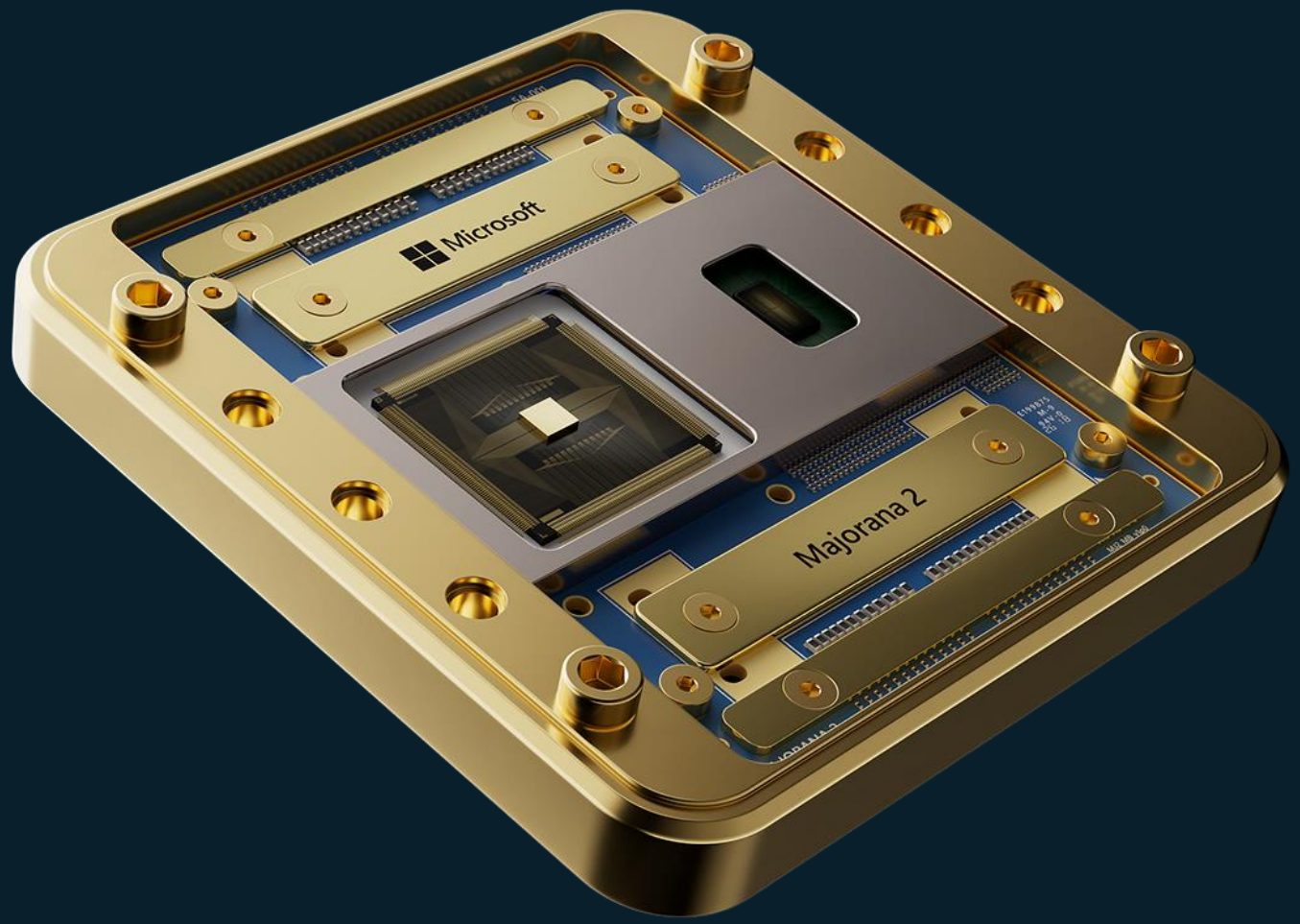
ML \ AI

Quantum algorithms for optimization

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Introducing The Majorana 2 Microsoft QPU?

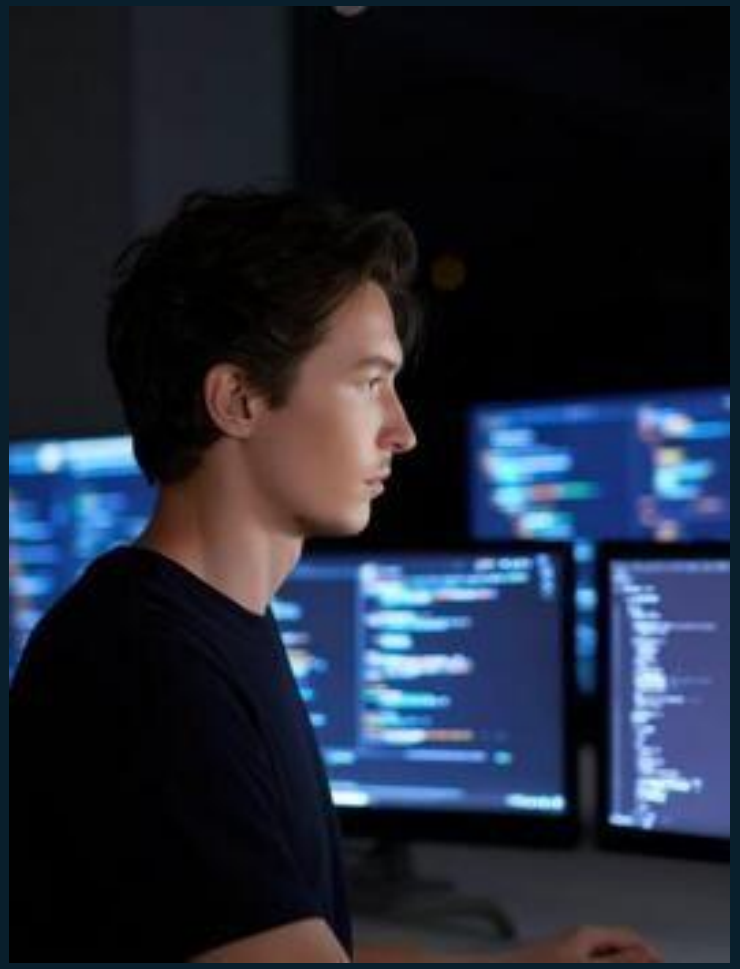




Microsoft's Majorana 2 chip: One Giant Leap for Mankind!

- Majorana 2 contains qubits that are 1,000x more reliable than those in Majorana 1 QPU.
- This rapid progress, enabled by AI, has cut our timeline in half for delivering a scalable quantum computer—now anticipated by 2029.
- Offers a pathway to fit a million qubits on a single chip that can fit in the palm of one's hand,

Developers it's Time to Start Now



Q# (Microsoft)	Quiskit (IBM)	Cirq (Google)
• Language • Q# • Description • Azure Quantum integrated	• Language • Python • Description • Most Popular, with Great docs	• Language • Python • Description • NISQ algorithms focus

Start Developing Today - Google

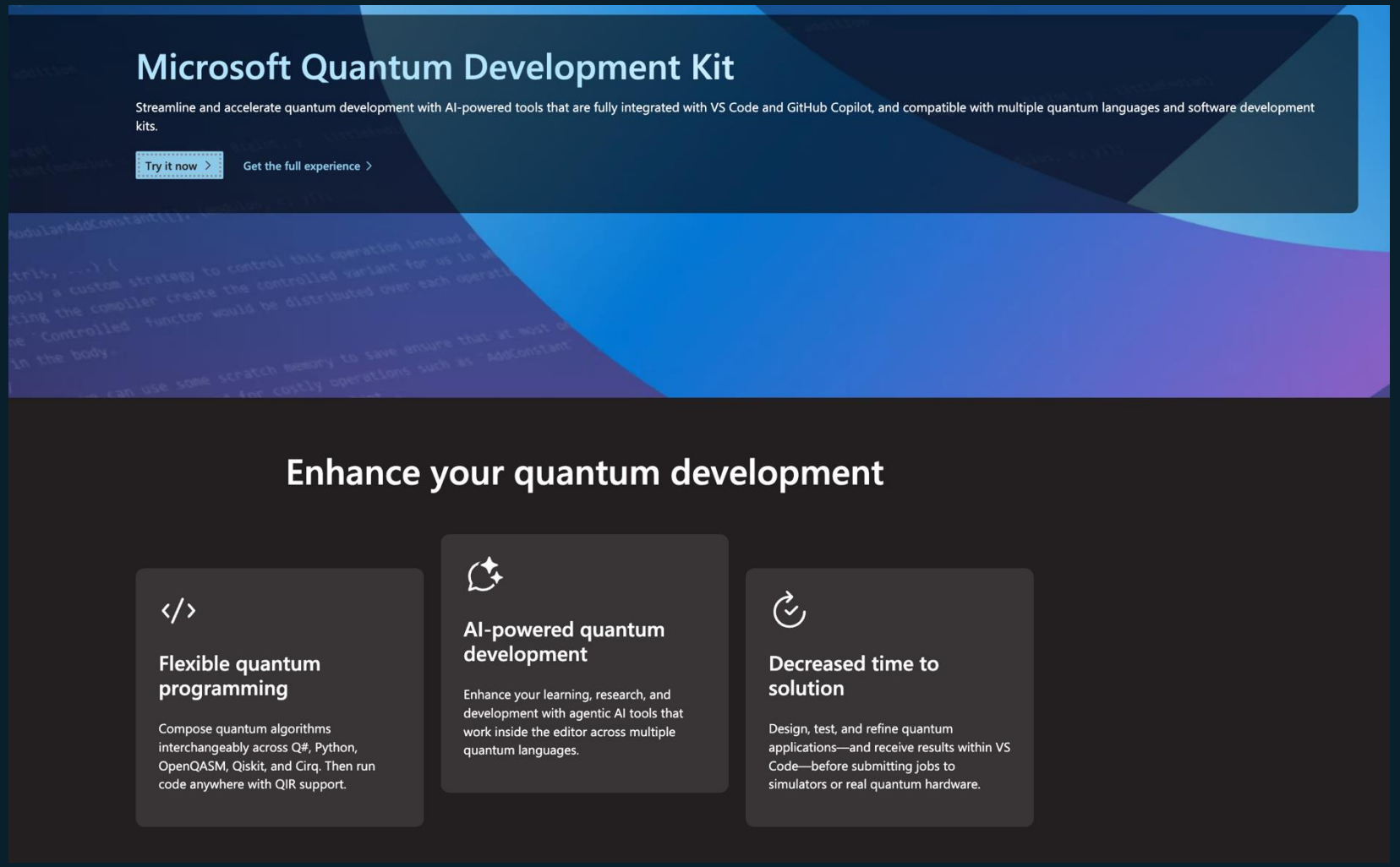
```
from qiskit import QuantumCircuit, QuantumRegister, ClassicalRegister
from qiskit_aer import AerSimulator

# 1. Create circuit
qc = QuantumCircuit(3, 3)
qc.h([0, 1, 2])  # Hadamard on each qubit
qc.measure([0, 1, 2], [0, 1, 2])  # Measure all

# 2. Run on simulator (FREE)
sim = AerSimulator()
job = sim.run(qc, shots=1000)
result = job.result()
print(result.get_counts())
```

https://quantumai.google/qsim

Start Developing Today - Microsoft



https://vscode.dev/quantum/playground/


Where will Quantum Take You?